

HAL Aerospace — LCA Tejas Mk1A Digital Twin

Framework & Technology Stack Architecture Specification | AEROTHON 2026

1. Executive Overview & System Architecture

The **HAL Aerospace Digital Twin Platform** is an enterprise-grade, full-stack mission control workstation engineered for real-time health monitoring, telemetry processing, and predictive maintenance of the **GE F404-IN20 turbofan engine** powering the **LCA Tejas Mk1A** fighter aircraft. Built for HAL Mission Control, the workstation unifies high-frequency thermodynamic simulation (Brayton cycle 0D/1D solver), 3-factor biometric access, interactive 3D graphics rendering, and explainable AI (XAI) diagnostics across fleet squadrons.

Layer	Component	Technology / Framework	Primary Responsibility
Frontend	User Interface	React 19 + Vite 6 + TypeScript 5	60 FPS Telemetry Dashboard, 3D Digital Twin Viewport, Fleet Operations
Frontend	State & Async Data	Zustand 5 + TanStack Query 5	Global UI state, telemetry store, server-side REST caching & revalidation
Frontend	Visualization & Graphics	ECharts 5.6 + Recharts + Three.js	Live gauge streaming, Weibull survival curves, SHAP waterfalls, SVG cutaways
Backend	REST API Server	FastAPI 0.140 + Uvicorn ASGI	Async REST endpoints, biometric auth verification, physics model execution
Backend	Persistence Layer	SQLAlchemy 2.0 + SQLite DB	Operator registry, mission logs, sortie replay history, audit trails
Backend	Security & Biometrics	PyJWT + Passlib + OpenCV + MediaPipe	JWT token creation, PBKDF2 hashing, 3-factor facial landmark extraction

2. Frontend Framework & Technology Stack Details

The presentation layer is built as a Single Page Application (SPA) using React 19 and Vite 6, providing instant render updates and modular UI components.

Technology / Library	Version	Category	Technical Usage in Platform
React	19.0.0	UI Core Framework	Concurrent rendering, functional components, custom telemetry hooks.
TypeScript	5.7.2	Language	Strict end-to-end type safety, engine sensor schemas, API interfaces.
Vite	6.1.0	Build Tool & Server	Lightning-fast HMR dev server and optimized production ESM bundler.
TailwindCSS	3.4.17	Styling & Design System	Utility-first styling, custom dark-mode aerospace cockpit color theme.
Zustand	5.0.3	Global State	Centralized state management for telemetry streams & selected squadron.

TanStack Query	5.66.0	Server State / Cache	Asynchronous API data fetching, query caching, and auto-refetching.
ECharts & ECharts React	5.6.0 / 3.0.2	Data Visualization	High-frequency telemetry charts (N1/N2 RPM, T4 temp), gauge clusters.
Recharts	2.15.1	Data Visualization	Weibull reliability curves, RUL predictions, SHAP waterfall charts.
Three.js & Unity WebGL	0.185.1 / 9.6.0	3D / CAD Viewport	Interactive GE F404 turbofan rendering, SVG thermal cutaway overlay.
Framer Motion	12.0.0	Animations	Micro-interactions, alert drawer transitions, diagnostic modal reveals.
React Router	7.1.5	Routing	Client-side view routing (Mission Control, Fleet, Diagnostics, Replay).
Lucide React	0.475.0	Icon System	Military-grade vector iconography for engine telemetry and status badges.
Zod	3.24.2	Schema Validation	Runtime data validation for telemetry payloads and operator auth credentials.
Playwright	1.62.0	Testing Framework	Automated End-to-End testing of critical mission control workflows.

3. Backend Framework & Technology Stack Details

The server application is built in Python 3.10+ using FastAPI, delivering high-performance asynchronous REST endpoints, database ORM bindings, and mathematical physics algorithms.

Technology / Library	Version	Category	Technical Usage in Platform
FastAPI	0.140.0	REST API Framework	High-performance asynchronous Python API with automatic OpenAPI / Swagger specs.
Uvicorn	0.51.0	ASGI Web Server	Asynchronous server gateway interface hosting FastAPI backend services.
SQLAlchemy	2.0.38	Database ORM	Object-relational mapping for operator data, login audits, and mission logs.
SQLite	Embedded	Database Engine	Low-latency relational storage (`hal_mission_control.db`) for mission registries.
PyJWT	2.13.0	Security & Auth	Generation and verification of cryptographically signed JWT access tokens.
Passlib	1.7.4	Security & Auth	Secure password hashing using `pbkdf2_sha256` with automated salt creation.
Pydantic	2.13.0	Data Validation	Data parsing, request model validation, and structured JSON responses.
NumPy	2.5.0	Numerical Computing	Vectorized math for thermodynamic Brayton cycle & biometric vector matching.
OpenCV (optional)	4.11.0	Computer Vision	Real-time webcam frame acquisition for military facial authentication.

MediaPipe (optional)	0.10.21	AI & Computer Vision	3D facial landmark mesh extraction for live 3-factor biometric verification.
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4. Core Engineering Modules & Technical Specifications

The platform integrates specialized defense aerospace engineering capabilities across key operational modules:

- **Military-Grade Biometric Auth:** 3-Factor authentication pipeline combining Operator ID, PBKDF2 hashed password, and live facial landmark verification (DEMO & REAL webcam modes supported).
- **Interactive Digital Twin Viewport:** High-precision SVG engine cutaway with interactive toggleable visual modes: Normal, X-Ray (structural stress), and Thermal (temperature gradient heatmaps).
- **Real-Time Telemetry Streaming Engine:** 60 FPS streaming processor tracking critical GE F404 metrics: N1 Low-Pressure Spool RPM, N2 High-Pressure Spool RPM, T4 Turbine Inlet Temperature (°C), Vibration (mm/s), Fuel Flow (kg/h), and Compressor Pressure Ratio.
- **AI Predictive Maintenance & RUL:** Weibull distribution hazard modeling, Anomaly Scoring engines, and Remaining Useful Life (RUL) forecasting based on cumulative flight hours.
- **Explainable AI (XAI) Diagnostics:** SHAP (SHapley Additive exPlanations) waterfall chart generation pinpointing top contributing factors behind flagged operational anomalies.
- **Physics-Informed Thermodynamics Solver:** 0D/1D Brayton cycle solver calculating real-time turbine inlet thermal loads, compressor operating maps, and aerodynamic surge margins.
- **Causal Fault Tree Analysis:** Interactive fault propagation tree tracking root-cause sequences (e.g., Fuel Injector Clogging → Hot Spot → Thermal Creep → Turbine Blade Micro-fracture).
- **Mission Replay & Historical Analytics:** Flight telemetry replay engine featuring 500 historical sorties with interactive surge event injection for operator training.

5. System Repository Directory Structure

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■■■■ backend/           # Python FastAPI Service
■  ■■■■ main.py         # REST API routes & app entry point
■  ■■■■ auth/           # JWT security & biometric engine
■  ■■■■ digital_twin/   # Brayton thermodynamic solver & AI models
■■■■ src/               # React 19 Frontend Source Code
■  ■■■■ components/     # Digital Twin, Charts, Telemetry gauges
■  ■■■■ store/          # Zustand state stores
■  ■■■■ types/          # TypeScript interface definitions
■■■■ hal_mission_control.db # SQLite Relational Database
■■■■ package.json       # Frontend Dependencies & Build Scripts
■■■■ vite.config.ts     # Vite ESM Config & Proxy Rules
```